TASK - VIRTUAL WORK LAB BY NOIVRN

Adverse Drug Reactions & Pharmacovigilance

A concise, visual study guide for B.Pharm students

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QUICK RECALL • EXAM TIPS • CLINICAL EXAMPLES

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01

Classification of Adverse Drug Reactions

ADR type	Mechanism / feature	Predictability
Pharmacological (Type A)	Exaggerated, dose-related response linked to the drug's known pharmacology.	Usually predictable; dose-dependent
Pathological (Type B-like)	Tissue or organ damage not explained by normal pharmacology.	Often unpredictable
Genotoxic	DNA damage that may cause mutagenic or carcinogenic effects.	May be delayed; exposure-related
Allergic (immunologic)	Antibody- or T-cell-mediated reaction after prior sensitisation.	Usually unpredictable; not dose-related
Idiosyncratic	Abnormal response associated with individual genetic or enzymatic variation.	Rare and unpredictable
Pharmacogenetic	Reaction linked to inherited variation in drug-metabolising genes.	May be predictable if genotype is known
Teratogenic	Exposure during pregnancy causes structural or functional fetal effects.	Depends on timing of exposure

CLINICAL EXAMPLES

Type A: hypoglycaemia after insulin overdose. Genotoxic: cyclophosphamide-associated secondary leukaemia. Idiosyncratic: aplastic anaemia associated with chloramphenicol. Pharmacogenetic: primaquine-associated haemolysis in G6PD deficiency.

REMEMBER

Type A reactions are commonly dose-related and predictable; unusual immune-mediated or idiosyncratic reactions may occur even when a drug is used correctly.

02

Hypersensitivity Reactions

Type	Also called	Mechanism	Example
Type I	Immediate / anaphylactic	IgE-mediated mast-cell and basophil degranulation.	Anaphylaxis with penicillin; allergic asthma
Type II	Cytotoxic	IgG or IgM binds cell-surface antigens; complement or cell injury follows.	Drug-induced haemolytic anaemia (e.g. methyldopa)
Type III	Immune complex	Antigen–antibody complexes deposit in tissues and activate complement.	Serum sickness; SLE-like reaction
Type IV	Delayed / cell-mediated	Sensitised T lymphocytes respond on re-exposure; antibodies are not the main mediator.	Contact dermatitis; tuberculin reaction

QUICK RECALL: ACID

I — Anaphylactic; II — Cytotoxic; III — Immune complex; IV — Delayed. Types I–III are generally antibody-mediated; Type IV is T-cell-mediated and commonly develops over 24–72 hours.

Timing is a useful clue, but real clinical reactions can vary. Suspected severe allergy or anaphylaxis needs urgent clinical assessment.

03

Drug Risk Categories in Pregnancy

Category	Traditional meaning	Examples from exam notes
A	Controlled human studies showed no fetal risk.	Folic acid; levothyroxine
B	Animal studies showed no risk but adequate human studies were lacking, or animal risk was not confirmed in humans.	Amoxicillin; metformin; paracetamol
C	Risk could not be ruled out; animal adverse effects or insufficient data.	Ciprofloxacin; fluconazole
D	Evidence of human fetal risk; benefit may outweigh risk in serious circumstances.	Phenytoin; ACE inhibitors; tetracyclines
X	Fetal risk clearly outweighs any possible benefit; contraindicated in pregnancy under this system.	Isotretinoin; thalidomide; methotrexate

EXAM TIP

Remember the direction: A is the lowest-risk category in this historical system; X is contraindicated. Thalidomide is a classic teratogen associated with phocomelia (severe limb defects).

CLINICAL CAUTION

The FDA replaced the old letter categories with narrative pregnancy and lactation labeling (PLLR). Do not use A/B/C/D/X alone to make current treatment decisions.

04

Causality Assessment Tools

Frameworksusedtoevaluatewhetheramedicinemayhavecaused an ADR

BRADFORD HILL CRITERIA

Nine viewpoints used to think about causation:
- Strength of association
- Consistency and specificity
- Temporality (cause precedes effect)
- Biological gradient / dose-response
- Plausibility and coherence
- Experiment and analogy

WHO–UMC CAUSALITY SCALE

Six categories: Certain, Probable/Likely, Possible, Unlikely, Conditional/Unclassified and Unassessable. Considers timing, alternative causes, and dechallenge/rechallenge information.

NARANJO ALGORITHM

A 10-question score-based tool (commonly –4 to +13). Typical interpretation: ≥9 definite; 5–8 probable; 1–4 possible; ≤0 doubtful.

PRACTICAL APPROACH

Review the timeline, known effects of the drug, dose and interactions, other possible causes, and what happened after stopping or restarting the medicine. Rechallenge should not be attempted when it could put the patient at risk.

05

Pharmacovigilance & Public Health

WHAT IS PHARMACOVIGILANCE?

The WHO describes pharmacovigilance as the science and activities relating to the detection, assessment, understanding and prevention of adverse effects or other medicine-related problems.

WHY IT MATTERS

- Identifies rare or delayed adverse effects after a medicine enters wider use
- Supports ongoing benefit–risk evaluation
- Encourages safer prescribing and rational medicine use
- Helps regulators communicate risks and update safety information
- Contributes to patient safety and public health

CORE IDEA

Clinical trials cannot identify every rare adverse effect. Reports from healthcare professionals and patients help build a broader picture of medicine safety once products are used in the community.

06

ADR Reporting in India

CDSCO•PharmacovigilanceProgrammeofIndia(PvPI) • ADR Monitoring Centres

PHARMACOVIGILANCE PROGRAMME OF INDIA (PvPI)

Launched in 2010. The programme is coordinated by the Indian Pharmacopoeia Commission (IPC), Ghaziabad, under the Ministry of Health and Family Welfare framework, with a network of ADR Monitoring Centres (AMCs) contributing safety reports.

HOW TO REPORT A SUSPECTED ADR

- Use the applicable CDSCO / PvPI suspected ADR reporting form or official reporting channel.
- Patients and healthcare professionals can report suspected reactions through available PvPI reporting routes, including the official app/portal where available.
- Provide patient details as required, suspected medicine(s), reaction description, dates and relevant medical history, outcome, and reporter contact details where requested.

GOOD REPORTING PRACTICE

Report a suspected reaction even when causality is not yet certain. Include clear dates, dose, indication, concomitant medicines, treatment given and outcome when known. Follow current official PvPI/CDSCO instructions for the applicable form or channel.